App Overview & Stack Context

This slide orients Claude Code. Fill in your details below.

Tech Stack

Frontend: Next.js (implemented)

Backend: python

ORM / Migrations: SQLAlchemy + Alembic

Database: SQlite

Deployment: Vercel

App Summary

An app used to explain to engineers & education professionals the 'why' and 'how' of a potential weeklong education unit. The 'Stories' feature contains slideshows on why this program might be good for students & professionals. The 'Design' feature shows the interactive tools which compose the subsystems of the windmill fleet.

User Roles / Personas

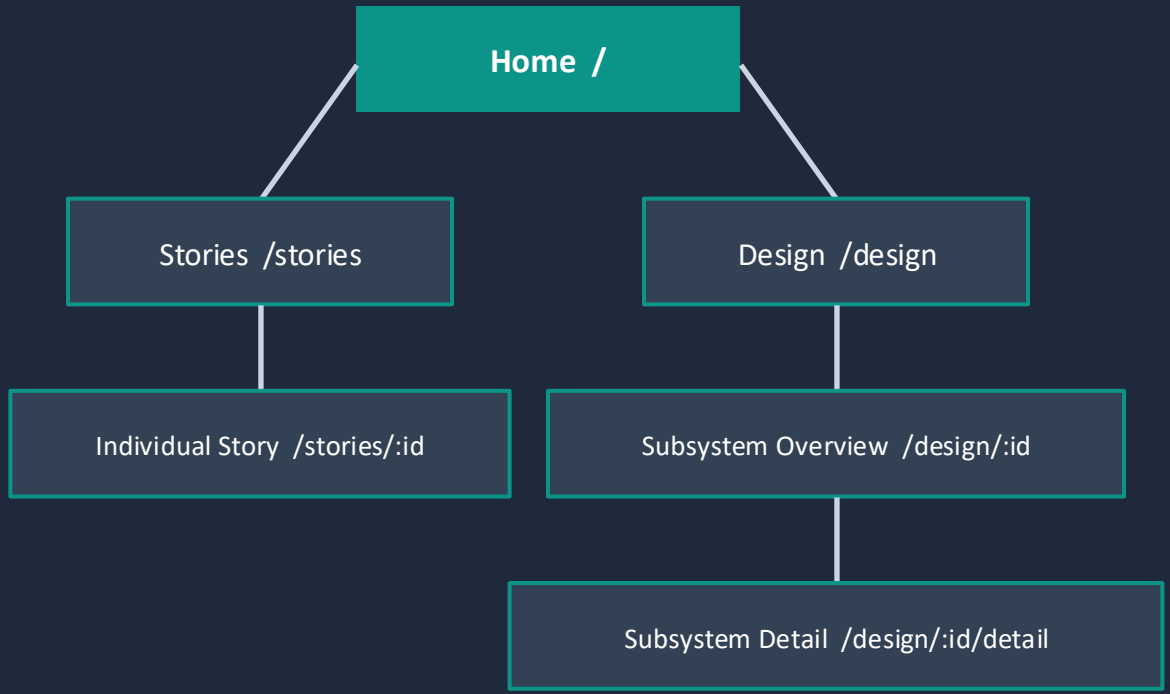
This app shall be public, no auth or concept of user.

Key Decisions Already Made

The existing home page and logic should not be overwritten. The HTTP API pattern has been established.

Navigation Map

Show how screens connect. Replace this with your own diagram or annotate below.

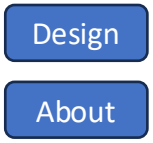


All of the lines represent two-way flows.

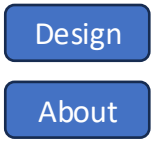
Home (Landing Page)

/

Desktop



Mobile



Purpose

The user changes the speed of the wind. The interactions & logic are already built in the page, in page.tsx.

Key Interactions

Users adjust a sliding bar at the bottom of the screen (not pictured, exists on main.tsx)

Data Shown

The data shown is already implemented in main.tsx.

Edge Cases & States

No loading. There are background illustrations. No concept of state necessary.

Stories

/stories



Purpose

User clicks one of two buttons

Key Interactions

Buttons lead to individual stories (/stories/:id)

Data Shown

Two buttons

Edge Cases & States

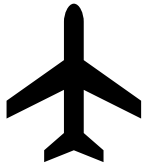
None

Individual Story (Template)

/stories/:id

Desktop

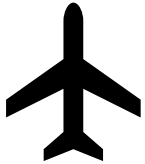
Top text



Bottom text

Mobile

Top text



Bottom text

Purpose

This page mimics Instagram story functionality. The number of bars at the top, correspond to the number of slides in this story. A slide is the unique combination of top text, bottom text, and image.

Key Interactions

Top text moves left to right, slowly. Bottom text moves right to left. Slow zoom in on whatever image is shown. Click (tap) advances slides. Click (tap) and hold stops the slide. Each slide has 6 seconds.

Data Shown

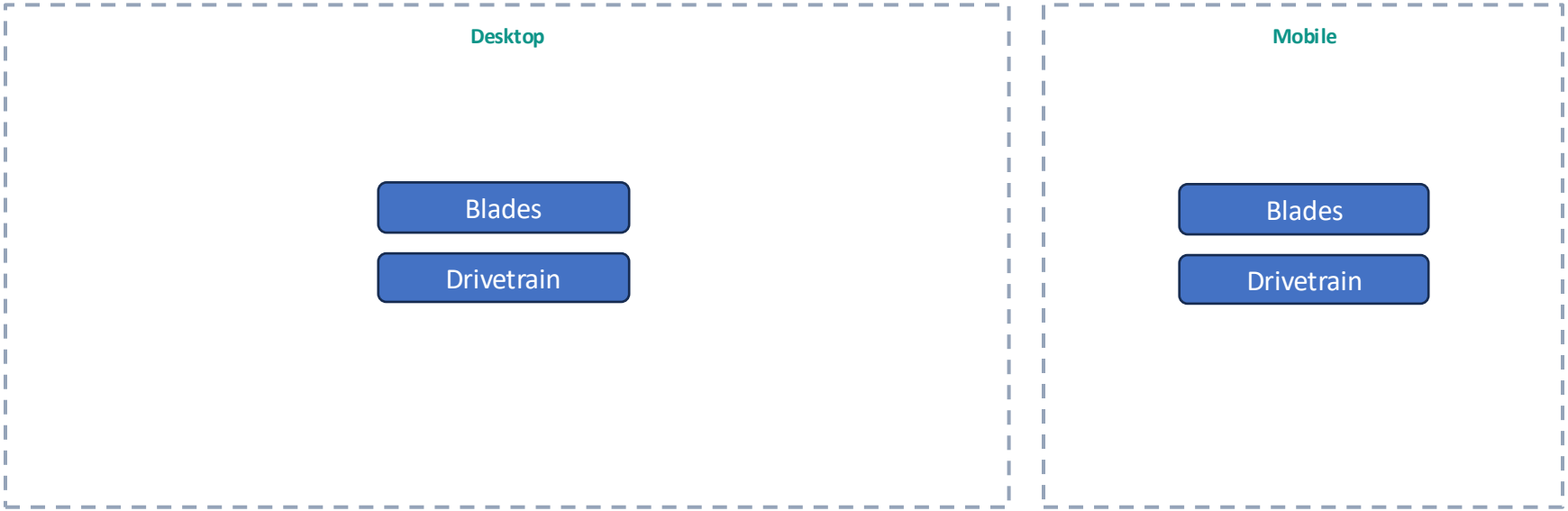
Each slide has its unique top text, bottom text, and image.

Edge Cases & States

The page will need to know which slide is currently showing, so it can go forward & back from that slide. When the slideshow ends, go 'back' to the Stories page.

Design

/design



Purpose

User clicks one of two buttons

Key Interactions

Buttons lead to individual stories (/design/:subsystemId)

Data Shown

Two buttons

Edge Cases & States

None

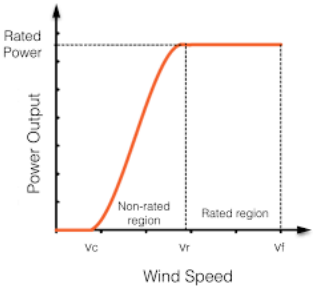
Subsystem Design Overview

/design/:subsystemId

Desktop

Back to Design | Blade Design





weight: 180
cost: 180
avg output: 180
efficiency: 180

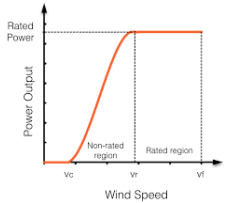


of blades: 3

Mobile

Back to Fleet | Blade Cross-Section





of blades: 3



weight: 180
cost: 180
avg output: 180
efficiency: 180

Purpose

The user inputs the number of blades and uses the slider to control the animation of blades & the dotted line on the chart to the right. Calculated values are output to the yellow boxes.

Key Interactions

Moving the slider
Inputting the number of blades, creates equally spaced blades on the turbine rotor.
Back to fleet goes to home. Blade design goes to

Data Shown

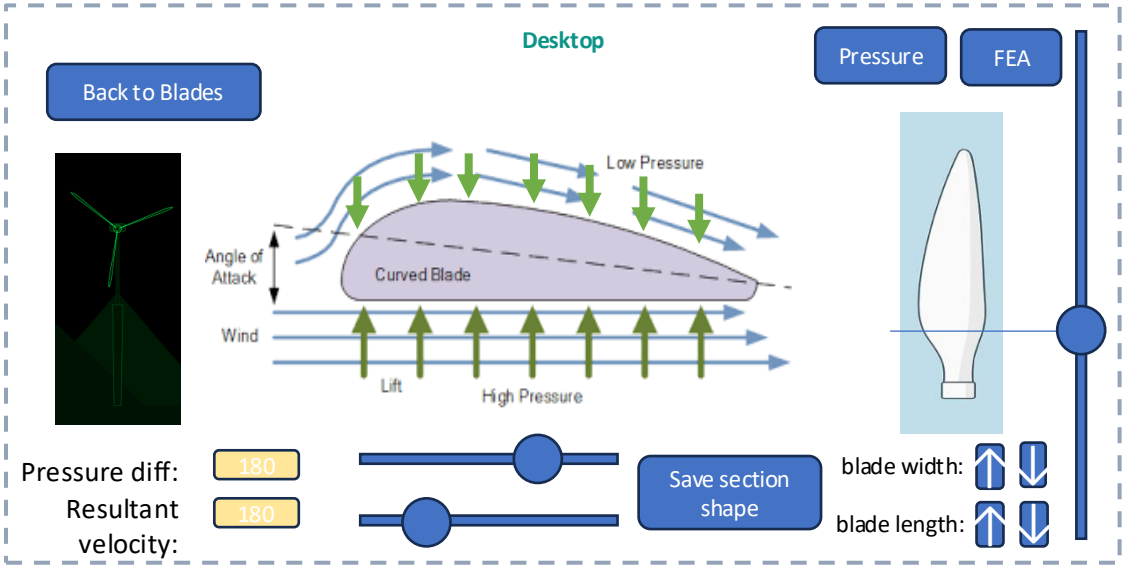
Text data is: weight, cost, avg output, efficiency. The chart is also interactive, x-axis is wind speed and y-axis is power generated.

Edge Cases & States

Number of blades drives the weight, cost, avg output, and efficiency output values. The slider changes the animation on the blades and on the chart.

Subsystem Design Detail

/design/:subsystemId/detail



Purpose

The purpose of this page is to allow a user to design a windmill's blades by editing the cross sectional area and shape. 10 discrete length locations of the blade can be modified by adjusting the slider on the right. The blade width and length can be adjusted. In the future (not now) cross sectional shape should be adjustable by click-and drag, but retain a base airfoil shape.

Key Interactions

On left: rotating windmill which displays overall blade shape. In middle: The arrows represent either airflow or pressure around a section of the airfoil. Their length corresponds to their magnitude. The user slides the two dials (angle of attack top, wind speed bottom). Save section shape, saves the shape of this cross section. On right: A single blade is shown. The slider controls where along the length of blade the visible cross section is. Blade width and blade length are controlled by arrow buttons.

Data Shown

Green arrows: Pressure gradient, caused by wind velocity, angle of attack, and shape of the blade. Larger arrow = more pressure.

Blue arrows: Animated wind flow. Equal magnitude. Velocity of wind flow based on wind velocity slider, angle of attack, and shape of blade. Not picture but should be shown: blade height, width cost.

Edge Cases & States

The value of the slider corresponds to a value on the blade's radial axis. This state must be known, as the values of angle of attack and shape of the cross section, are saved for each length value. The windmill animation (far left) must know the overall blade geometry (shown on right)

Shared Patterns & Conventions

Document global decisions so Claude Code doesn't reinvent them.

Navigation & Routing

Routing buttons will have subtle arrow shapes.

API Response Format

API responds with JSON.

UI Components & Styling

The UI should be OK for colorblind people. The styles can hopefully be modelled after the latest iOS UI.

Content & Animations

The 'Individual Stories' feature should mimic the user action of Instagram stories. The app is small enough that static assets can be used.